# Final Project Report: Secure Enterprise Network Design

## 1. Executive Summary

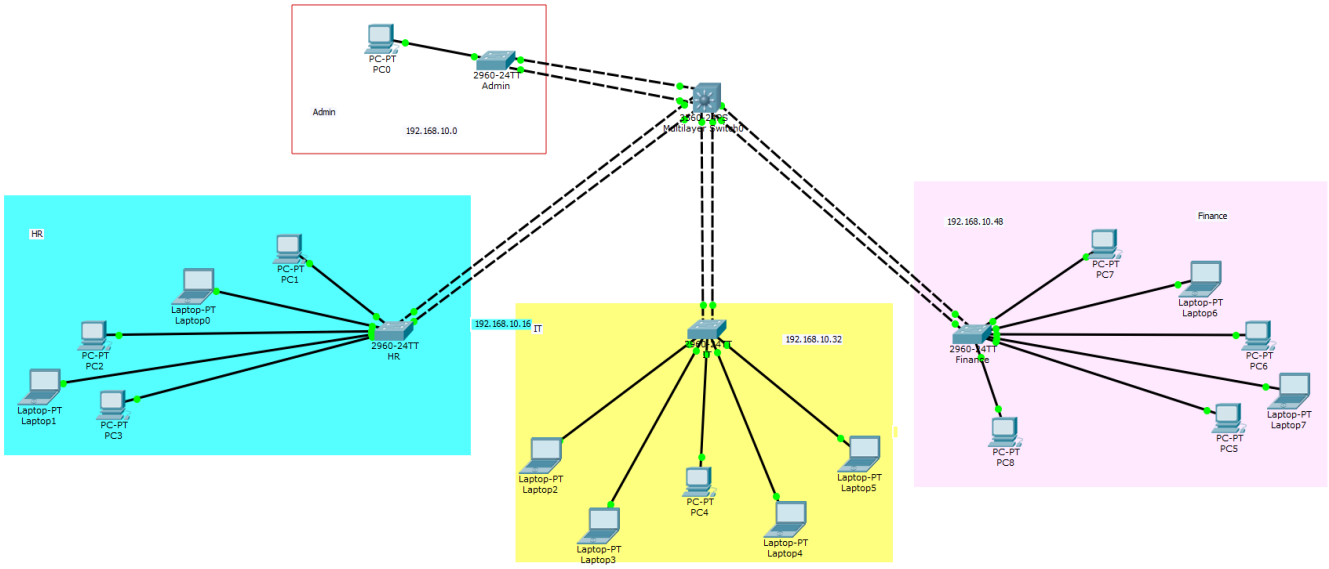
This project successfully implemented a secure, redundant network for a small enterprise consisting of four departments: Admin, HR, IT, and Finance. By utilizing a Layer 3 switch for Inter-VLAN routing **(Admin can ping to any other VLAN but departments PC can only communicate with their same department only)**, implementing EtherChannel for link redundancy, applying ACLs for security and implementing VLSM, we created a scalable environment that minimizes downtime and prevents unauthorized data access.

## 2. Network Design & VLSM Implementation

We used **VLSM (Variable Length Subnet Masking)** to prevent IP address wastage. Each department was allocated a block of 16 addresses (/28), allowing for 10 usable hosts (5 current + 5 future growth).

- **Subnetting Table:**

| |
|---|
| ○ **Admin (VLAN 10):** 192.168.10.0/28 (Gateway: .1). |
| ○ **HR (VLAN 20):** 192.168.10.16/28 (Gateway: .17). |
| ○ **IT (VLAN 30):** 192.168.10.32/28 (Gateway: .33). |
| ○ **Finance (VLAN 40):** 192.168.10.48/28 (Gateway: .49). |




## 3. Methodology (Implementation Steps)

1. **VLAN Segmentation:** Created 4 VLANs to logically separate departmental traffic.
2. **Redundancy:** Configured **EtherChannel** (LACP/Static) between the Access Switches and the Core Switch using dual links to ensure the network stays up if one cable fails.
3. **Routing:** Enabled ip routing on the Multilayer switch to allow the Admin department to communicate with all other sectors.
4. **Security (ACLs):** Applied Extended Access Control Lists on the SVI interfaces to **restrict** HR, IT, and Finance from talking to each other while still allowing Admin to manage them.

Multilayer Switch0

Physical | Config | CLI

IOS Command Line Interface

```
FastEthernet0/23       unassigned      YES unset  down                  down
FastEthernet0/24       unassigned      YES unset  down                  down
GigabitEthernet0/1     unassigned      YES unset  up                    up
GigabitEthernet0/2     unassigned      YES unset  up                    up
Vlan1                  unassigned      YES unset  administratively down down
Vlan10                 192.168.10.1    YES manual up                    up
Vlan20                 192.168.10.17   YES manual up                    up
Vlan30                 192.168.10.33   YES manual up                    up
Vlan40                 192.168.10.49   YES manual up                    up
Port-channel 1         unassigned      YES unset  up                    up
Port-channel 2         unassigned      YES unset  up                    up
Port-channel 3         unassigned      YES unset  up                    up
Port-channel 4         unassigned      YES unset  up                    up
Switch#
```

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```
VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24
10   Admin                            active
20   HR                               active
30   IT                               active
40   Finance                          active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

Vlans

```
Switch#show access-lists
Extended IP access list SECURE_HR
    10 permit ip 192.168.10.16 0.0.0.15 192.168.10.0 0.0.0.15 (2 match(es))
    20 deny ip 192.168.10.16 0.0.0.15 192.168.10.32 0.0.0.15 (4 match(es))
    30 deny ip 192.168.10.16 0.0.0.15 192.168.10.48 0.0.0.15 (1 match(es))
    40 permit ip any any
Extended IP access list SECURE_IT
    10 permit ip 192.168.10.32 0.0.0.15 192.168.10.0 0.0.0.15 (9 match(es))
    20 deny ip 192.168.10.32 0.0.0.15 192.168.10.16 0.0.0.15
    30 deny ip 192.168.10.32 0.0.0.15 192.168.10.48 0.0.0.15
    40 permit ip any any
Extended IP access list SECURE_FINANCE
    10 permit ip 192.168.10.48 0.0.0.15 192.168.10.0 0.0.0.15 (2 match(es))
    20 deny ip 192.168.10.48 0.0.0.15 192.168.10.16 0.0.0.15 (1 match(es))
    30 deny ip 192.168.10.48 0.0.0.15 192.168.10.32 0.0.0.15 (1 match(es))
    40 permit ip any any
```

Access Lists

```
Switch#sh etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 4
Number of aggregators:           4

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-------------------------

1      Po1(SU)           -      Gig0/1(P) Gig0/2(P)
2      Po2(SU)           -      Fa0/1(P) Fa0/2(P)
3      Po3(SU)           -      Fa0/3(P) Fa0/4(P)
4      Po4(SU)           -      Fa0/5(P) Fa0/6(P)
```

Etherchannel

## 4. Results & Testing

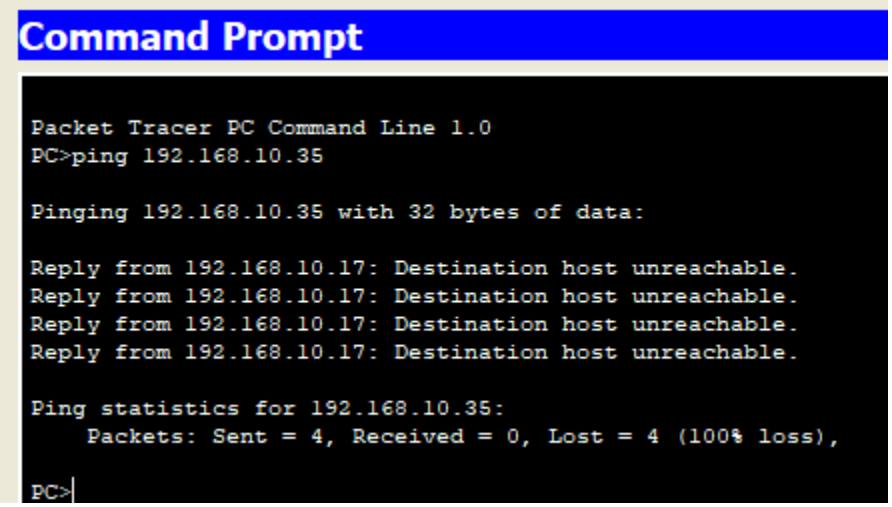
- **Inter-VLAN Communication:** Verified that the Admin PC could successfully ping all other departments.
- **Security Effectiveness:** Pings between HR and Finance were "Destination Prohibited," proving the ACLs successfully blocked unauthorized lateral movement.
- **Redundancy Test:** We manually deleted one link in an EtherChannel bundle. Traffic continued to flow through the second link without interruption, confirming high availability.

***Admin to IT,***

```
PC>ping 192.168.10.35

Pinging 192.168.10.35 with 32 bytes of data:

Reply from 192.168.10.35: bytes=32 time=0ms TTL=127
Reply from 192.168.10.35: bytes=32 time=0ms TTL=127
Reply from 192.168.10.35: bytes=32 time=0ms TTL=127
Reply from 192.168.10.35: bytes=32 time=0ms TTL=127

Ping statistics for 192.168.10.35:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Successful

*"Failed" ping from HR to IT*




Failed

## 5. Challenges & Conclusion

The main challenge was correctly calculating the VLSM boundaries to ensure subnets did not overlap while leaving room for the required 10 PCs. Additionally, ensuring the EtherChannel was properly negotiated required matching speed and duplex settings on both ends.

In conclusion, the project meets all objectives of the proposal by providing a cost-effective, secure, and resilient network foundation for the enterprise.